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A novel need based free channel selection scheme for cooperative CRN using EFAHP-TOPSIS

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ABSTRACT

Emerging wireless applications needs more spectrums space to tackle the rapid growth of users. Outside of radio band in spectrum will be hard to utilize for data transmissions in terms of hardware requirement. Cognitive radio research considers the Current Spectrum underutilization and provides a better model for next-generation wireless environment. Since, the cognitive radio and its policies violate the static spectrum allocation i.e., Current Wireless networks policy, many challenges are in front of us to accomplish a better cognitive radio wireless environment. One of the major challenges is secure transmission and optimal free channel selection for un-interrupt data transmission. In this article, we have proposed an efficient free-channel selection scheme for improving QoS of Cooperative Cognitive Radio Networks. The proposed model has been derived as an integrated approach by using successful MCDM techniques such as EFAHP and TOPSIS. Also, we have proved the optimality of the proposed integrated technique with other parallel channel selection techniques with various scenarios.

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1. Introduction

Cognitive Radio Network (CRN) is one of the standard wireless network communications which utilizes un-used spectrum bands from licensed users with permission. Licensed users are also named as Primary users (PU) and users who are accessing available un-used spectrum named as Cognitive radio (CR) users or secondary users (SU). Since the increasing demand for spectrum and rapid development of the wireless application, the underutilized spectrum needs to be used efficiently and using new spectrum outside conventional radio band require un-imaginable hardware. So CRN is one of the hopeful methods to match the spectrum requirement of forthcoming generation (Taherpour et al., 2017; Jayakumar et al., 2019).

At a particular time and specific geographic location, PU may not utilize allotted spectrum for its service then that will be treated as spectrum hole or white space or free channel since spectrum is a precious resource it has to be accessed efficiently. For increasing

the consumption ratio of spectrum allotted to licensed users, secondary unlicensed users should dynamically access the spectrum holes that are not currently occupied by the primary user (Ali and Hamouda, 2017).

1.1. Related works in CRN spectrum management

Since CRN has unique characteristics on their transmission based on dynamic spectrum, concentration on managing spectrum should be more. Spectrum is a costlier resource, the underutilized spectrum is reused by using CRN alone will not increase in the utilization of spectrum. Also, efficient spectrum management schemes should get implemented to utilize available opportunities efficiently (de Almeida Filho et al., 2018; Singh et al., 2016). As in Fig. 1, spectrum sensing will done by centralized stations and spectrum decision can be done by either centralized or distributed to participating CR's in CRN. It is level by level process, spectrum sensing will be in the first level to finalize a list of spectrum opportunities in the radio environment. Later, better option or choice to utilize will be decided at the spectrum decision level. These two key areas are discussed as follows,

1.1.1. Spectrum sensing

List of various spectrum opportunities or spectrum hole surrounded by CR device will capture by using the spectrum sensing

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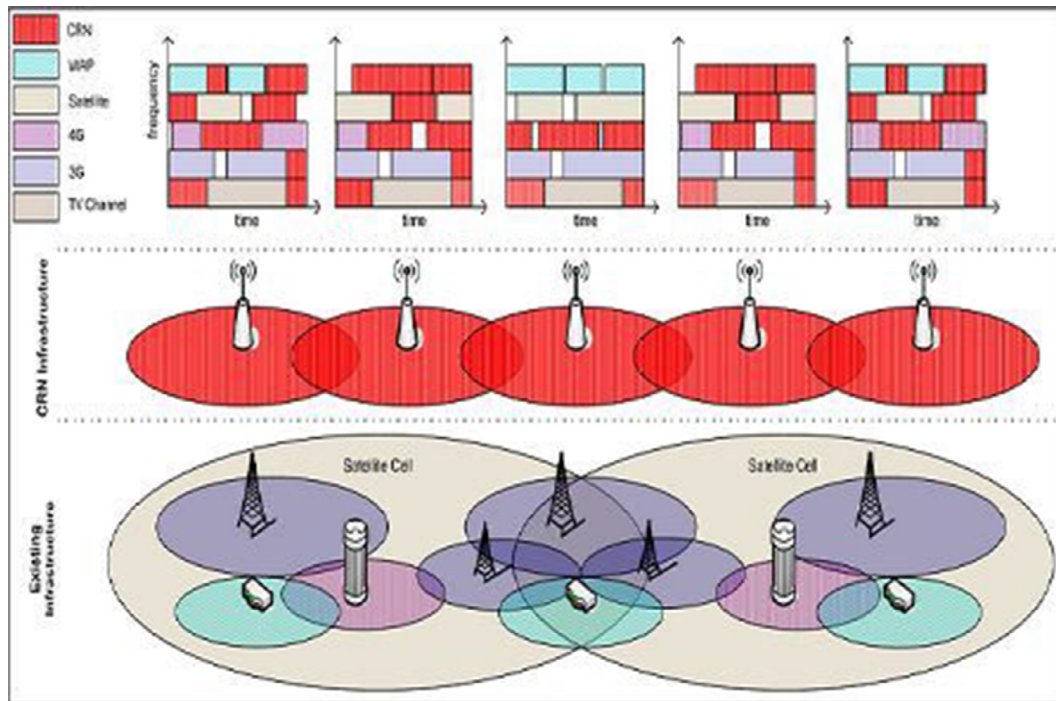


Fig. 1. Cooperative CRN.

task (Akyildiz et al., 2006). Practical lab experiments can be done by using CR devices released by Ettus industries which are used by most of the researchers such as USRP N210, USRPN310, etc., sensing range or capability of these CR devices is from 600 MHz to 6 GHz. The following methods commonly they are using in those devices,

- Energy Detection (Liu et al., 2012; Kumar et al., 2016)
- Cyclostationary detection (Annapurna, 2016)
- Auto-correlation (Naraghi-Pour and Ikuma, 2010)
- Wavelet method (Salahdine, 2017)
- Matched filter (Dannana and Chapa, 2018)

Energy detection and matched filter sensing techniques will be initialized with the threshold energy level for primary users. If the primary user uses the allotted spectrum then detected energy exceeds the threshold level, otherwise lower level of detected energy indicates availability of that band. Other sensing methods like Cyclostationary Detection, Auto-correlation, and Wavelet method are applying multiple factors to detect availability. Probability of availability will be calculated in the latter group of techniques. In (Salahdine, 2017), the author states energy detection technique outperforms well than other sensing techniques.

In this article, energy detection based sensing results are suggested and proposed sensing suggesting only better cooperative network model and effective task management at spectrum sensing phase. The following sections are basic studies have completed before finalizing the proposed channel management model. In that way, firstly sensing environments and its types have been analyzed here as follows,

- Free-use: some of the geographical locations like the middle of the Ocean, dense forest, tunnel and mine most of the spectrum will be free. So, without much effort spectrum decision can be taken for data transmission (Costa et al., 2013).

- Shared-use: metro cities, towns and capitals cities of countries most of the spectrum bands will be occupied and villages also particular cellular, terrestrial and satellite bands has been occupied by various service providers. In the future, for implementing new wireless service CRN can be implemented and underutilized spectrum can be used (Sabbah et al., 2018).

In spectrum sensing task, CRN architecture and structure will be a major role. Network structure can be majorly classified into two types: cooperative and Non-cooperative CRN. In the non-cooperative model, each CR node should take responsibility to detect the availability of the surrounding radio environment for finding a free channel for data transmission. But which could lead to interference, noise uncertainty, fading, shadowing and huge energy loss for sensing wide spectrum. Quality of decided channel cannot be assured with individual sensing and decision. So, most of the existing research carried out with cooperative sensing and decision CRN models. Cooperative sensing can solve the hidden primary user problem and can decrease sensing time. In addition, it decreases the probability of miss-detections and the probability of false alarms considerably. One of the major challenges of cooperative sensing is to synchronize the channel availability or activity information to all other CR participating in the channel decision model. (Axler, 1995; Zyoud et al., 2017). Cooperative sensing can be classified into two types: centralized or distributed (Lorenzo et al., 2018).

- **Centralized Sensing:** Though cooperation among participating CR nodes in CRN, centralized Fusion Center will acting sensing mode for the entire network. This is possible when the centralized node is fixed and capable to keep the charge for handling wideband spectrum sensing. But, most of the scenarios, all CRN nodes are handheld and limited charged devices. Channel decision can be centralized or distributed along with a central sensing model. Compared to the sensing task, channel decision

is a less computational and simpler task, so centralized sensing and decision will be better than central sensing and distributed decision as in Fig. 2 (Yan, 2002).

- **Distributed Sensing:** In the case of distributed sensing, CR nodes can be sensed individually and sensed results will be shared among themselves to take global decision across CRN. Distributed sensing is more advantageous in the sense that there is no need for a backbone infrastructure (Cabric et al., 2004). But, if any status changes in channels that should be communicated to all other distributed CR's using Efficient Distributed Algorithms. Synchronization, Timely updates are the factors influenced majorly in Distributed Sensing schemes (Changqing Luo et al., 2011).

1.1.2. Spectrum decision

In CRN, decision-making problems are needed in three situations. 1) deciding right list channels for sensing for avoiding misdetection of free channels, 2) deciding CR nodes for using them for making channel decision and 3) optimal channel decision among the list of sensed available free channels for long time transmission (Yi and Cai, 2018). In this research based on primary assumption, each and every channel are to be sensed under channel sensing network. one of the reasons for cooperative network suggested to implement in existing work is reduce the sensing load. Workload has been distributed over all participating CR nodes. The first type of decision-making problem need not be emphasized because of the cooperative nature of sensing got familiar and know to be a successful model among others. The second scenario to take a decision for fixing CR nodes for making channel decision, if multiple CR nodes are decided as decision making nodes then global optimal channel decision will not be possible. A local decision from some of the decision nodes (DN) may not allocate some best suitable free channels to the needed CR users (Sun et al., 2018; Bhargavi and Murthy, 2010; Muhammed Enes Bayrakdar, 2016). So, achieving global optimum channel decision, centralized channel decision can be suggested (See Fig. 3).

2. Construction of efficient free-channel selection scheme

As mentioned earlier, an integrated approach used to develop this novel channel selection scheme. For constructing channel selection scheme, required network model and channel fusion and filtration have to be carried out as primary steps. Since each channel having different capabilities and properties, it may fit into different types of data transmission. Example, for video stream data transmission, channel capacity should be high for transmit-

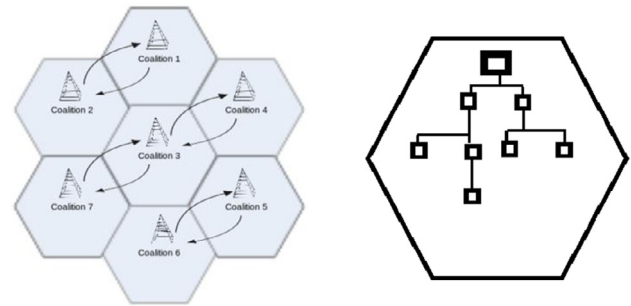


Fig. 3. (a). Multi-hop Cooperative CRN, (b). Cooperative CRN Hierarchical cluster.

ting channels but it may not require for text data. So, one-side number of channels with different capacities and another side different ranges of requirement require optimal channel selection or decision scheme to allocate channels dynamically when ongoing data transmission (Santiago et al., 2015; Eppstein, 2000). Ultimately objective for this work is to match the available channels with the right data transmission. The following steps are to be carried out to accomplish the proposed channel selection model,

- Network Formation
- Local Channel Sensing
- Channel fusion and Channel Decision

2.1. Network Formation

Nowadays cooperative cognitive radio networking (CCRN) has got noticeable attention, where cooperative networking is influenced to create a win-win situation for both Primary users and secondary users (Jia et al., 2016; Dyer et al., 1992). So, here we considered the cooperative environment as a platform for our proposed free-channel selection problem. Even though numerous ways to create cooperation between network nodes, clustering mechanisms have some considerable advantage over others. We have compared k-mean, k-medoid, and multilevel hierarchical clustering methods and its performance for CCRN. In (Ali and Hamouda, 2017), we have proved that multilevel hierarchical clustering is a suitable method to form a cluster of the secondary user for CCRN and channel sensing task split operations. A hierarchical cluster can be done by ways: first, agglomerative is a bottom-up clustering method, iteratively nodes will be merged into cluster based on its distance from Cluster Head (CH). Here clusters have sub-clusters which in turn have sub-clusters. But for a divisive method of hierarchical clustering, all participating CR will be considered as a single cluster, and then clusters are recursively split to yield a good clustering. We have fixed multilevel hierarchical clustering using the agglomerative method, it is better suitable network structure for cooperative sensing and sensing task split in our project (Biscarri, 2017; Franti et al., 2006).

CH plays a vital role in our project because from initial task split operation for participating nodes in the cluster, channel fusion to channel decision (Shahen Shah, 2014; Ganesan and Y. Li, 2005). Influence of CH is more important. Example, if we create a new CCRN with 100 nodes, Allotted sensing Bandwidth 200 MHz, each channel size 25 KHz then for each time 8000 number of channel sensing task has to be carried out for collecting availability of each channel in the given 200 MHz. It is difficult to perform by a single CH or participating nodes, so based on some quality parameters (Squared Euclidian distance, Reserved Power, willingness and failure rate) of each node, we have split taken 8000 sensing task into small sub-tasks and allocate to each node. This type of allocation

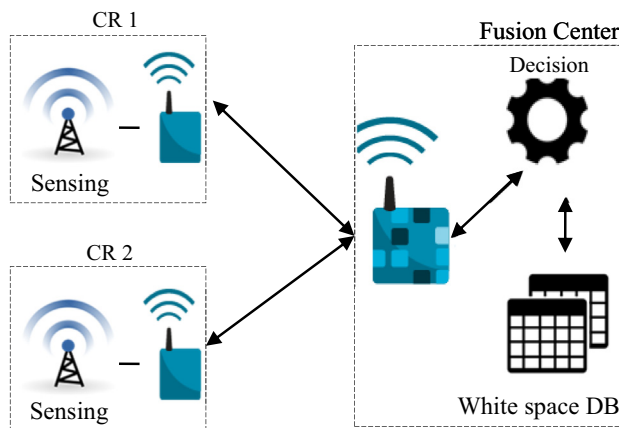


Fig. 2. Distributed Sensing – Centralized Decision.

can be carried out efficiently using agglomerative based multilevel clustering (See Fig. 4).

For each node addition in agglomerative method follows the Auction Process (Auction – I). It considers the following Node Selection Function (SF). Initially, the clustering process will pre-calculate the threshold level of accepting the participating for a cluster. That threshold level may vary based on our objective of network creation, available node count and its capabilities.

$$SF(n) = E_{sd} + T_{rp} \quad (1)$$

where,

E_{sd} – Squared Euclidian distance

T_{rp} – Total Reserved Power

2.1.1. Bidder's winning condition

Based on the availability of nodes present in the CRN or based on the free channel need, threshold value for filtering CR node to construct cooperative CRN can be fixed. When threshold value increases, number of nodes with higher capabilities will be selected for CRN, on the other hand if threshold value decreases obviously number of nodes will be increased even with lower capability (Ganesan and Li, 2005; Ghasemi and Sousa, 2007). The basic representation of this winning condition specified in Eq. (2).

$$SF(bidder(n)) > \text{Threshold}(\min) \quad (2)$$

2.2. Local free-channel sensing

Once each node in the cluster allocated with sensing task, local channel sensing task will be initiated. Check the status of each channel, whether available or occupied by primary user.

Result of each node's will either "ON" or "OFF". If, the status is "OFF" as represented in Fig. 5, from this we can conclude that for that channel can be used for CR node transmission or else we should avoid that channel for transmission (González-Rodríguez et al., 2012; Tian et al., 2018).

Here, we have enhanced exist local channel sensing scheme; earlier they considered only availability of each channel for calculating CAV (channel availability vector). In this, we considered the usage pattern of each channel in past history and priority values of current occupied user (Muhammed Enes Bayrakdar, 2017). If Best Order Factor (i.e., Lengthy free slots) is high, it will increase decid-

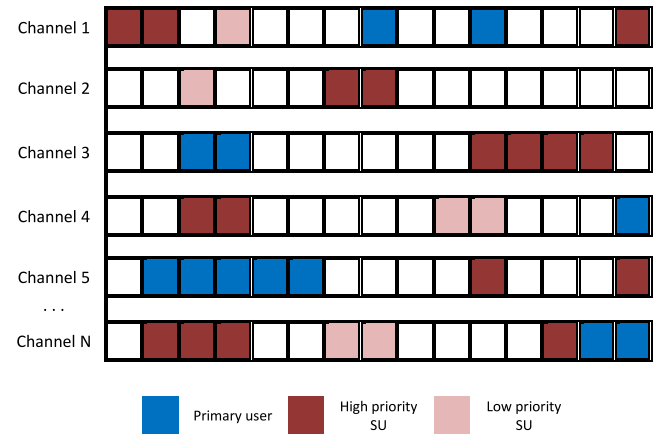


Fig. 5. Historical availability or occupancy details about N number of channels in CRN.

ing factor, of course T_{off} considered as a availability factor as like as exiting scheme (He et al., 2010; Ghazzai et al., 2017). Some of the sample values of CAV represented in Table 1.

Channel Availability Vector (CAV)

$$CAV_{\alpha} = \frac{\begin{matrix} C - \text{participatedno.ofCR} \\ L - \text{participatedno.ofPR} \\ BOF - \text{Bestorderfactor} \\ N - \text{no.ofchannels} \\ Slp - \text{slotsoccupiedbylowpriorityuser} \\ Shp - \text{slotsoccupiedbyhighpriorityuser} \\ Toff - \text{numberofunusedtimeslots} \end{matrix}}{\quad} \quad (3)$$

After calculating CAV value for each channel from each node, will be forwarded to CH. If number of channels is more, then the number of comparison computation in CH will be more. It may degrade the performance of CH. We can apply channel level Auction (Auction-II) for avoiding unnecessary channels for channel decision at CH. For each channel, we can apply threshold comparison for moving that channel to CH (Huang and Yuan, 2018; Liao et al., 2015; Cao et al., 2017).

$$CAV(ch_ID) > CAV_Threshold(\min) \quad (4)$$

2.3. Channel fusion and channel decision

Filtered Channels from each node will be reached at CH, Using simple data fusion we can easily create a master table. Instead of simple fusion technique, some of them are using AND and OR rule-based fusion, for that each node sense the same set of channel all the results will be collected at CH, then fusion will give each channel result as available or not. Here, we can perform only minimal set of channel sensing task, because all our resource nodes trying to collect same result. That is why, we selected simple data

Table 1
Sample: CAV values sensed by CR.

Channel	Current user	CAV
600 MHz	PR034	0.41
601 MHz	PR045	0.52
603.5 MHz	Free	0.54
665.6 MHz	PR024	0.56
715 MHz	CR067	0.46
.....		
Channel N	Free	0.43

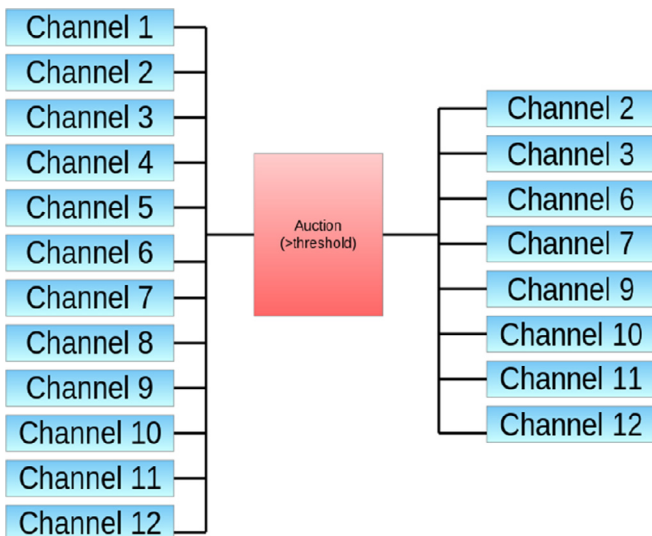


Fig. 4. Example illustrations of Filtering of Quality channels in the first level.

Table 1.1
Multi-criteria decision models.

AHP	Analytical Hierarchical Processing (Business and Technical) (Kakalou et al., 2017)
TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution (Ranking) (Loganathan and Sukumar, 2012)
ELECTRAE	ELimination and Choice Expressing Reality for (energy, economics, environmental, water management, transportation problems) (Jia and Shen, 2008)
PROMETHEE	Preference Ranking Organization Method for Enrichment Evaluations (partial ranking of the alternatives and the PROMETHEE II for complete ranking of the alternatives (Zhang et al., 2008)
VIKOR	VlseKriterijumska Optimizacija I Kompromisno Resenje (Kang et al., 2018) (Loganathan and Janakiraman, 2016)
SAW	Simple Additive Weight (Jayakumar Loganathan et al., 2016)

fusion, here different set of channel for each node. Here, we have chosen famous operation research domain solution named Multi-Criteria Decision Making (MCDM) (Hwang and Yoon, 1981; Huang et al., 2013).

a. Multi-Criteria Decision Making

It is sub-domain of operation research and classical decision problems in various engineering domain and management has been solved under various criteria. In Table 1.1, we have listed some of the most successful classical Multi-Criteria Decision Making models with its feature briefly.

b. Channel Decision by Channel Ranking

Ordering the channels as per their weight from decision-making models, and decide the channel which is in first rank when SU request for transmission. Here we used the combination of EFAHP and TOPSIS as decision-making schemes (Gow et al., 2012; Li et al., 2018). Both techniques are famous classical decision-making techniques in management, economics and network routing, detailed analysis has been discussed in chapter 3.

3. Proposed free channel selection using efahp-topsis

So far channel decision schemes considered channel availability factor alone to judge about quality. But, the requirement or necessity of wireless application may vary from time to time. In our proposed work along with the availability of channel some other performance factors also considered. Even availability of channel also not taken as just a percentage of slots occupied by PR, we have applied intelligently to detect the best order of availability as in chapter 3.1.1. Usually, history-based calculations are fully depending on the traffic pattern in the sense of “ON” or “OFF” in the time slot based channel transmission. That will be converted into $N \times T$ matrix and each row treated as the historical value of CR's, where N is a number of channels and T is a number of historical time slots considered for channel decision. Using this type of additional calculation of availability factor, computation time may increase a bit but the quality of results leads to better channel selection. The simulated experiment also shows a better result for our proposed work. In Fig. 5, sample channel usage representation has been displayed with 5 channels. Even secondary users again can be classified into the low priority and high priority users to occupy the primary user's channel.

Using our proposed method, we have considered six performance parameters of channels and collected past historical data for T time slots as in Table 1.2. Initially, one of the performance parameter Channel Availability Vector (CAV) has to pre-processed as per Eq. (6) and other factors have to be collected for

Table 1.2
Considered parameters for channel selection technique.

Parameters influenced in CRN performance
Channel Availability Vector CAV (%)
Bandwidth (MHz)
Probability of packet loss (PL) (%)
Signal power (dBm)
SNR(dB)
Maximum rate of transfer (bps)

T time slots. Then, we should fix our requirement like data service type is text, audio or video. Priority order of performance factors will be defined for each service. Finally, the dataset for T time slots and priority order of required data service will be given as input for AHP-TOPSIS which is an integrated operational research technique to rank the channels based on ongoing data transmission. An experimental result in Chapter 4 proves the performance of proposed work in multiple scenarios.

3.1. Preprocess parameters for channel selection

3.1.1. Channel availability vector CAV (%)

One of the primitive requirement for any kind of data transmission is channel should be available for completing transmission with fewer handoffs. As we mentioned earlier primary user activities has to be recorded by using two binary states “ON/OFF”. Frequency of arrival of the primary user at the particular time always unpredictable and random, but overall availability ratio and nature of the usage of channel can be collected for further channel decision. In this article, we have fixed sensing at local CR node and decision at FC, so all CR's in CRN has to forward the sensed channel usage details to FC. The following equations clearly represent, channel availability vector (CAV),

- Channel availability:

$$T_{\text{off}} = \sum_{\text{currentch}=i}^N \begin{cases} 1, & \text{if ch}(i) \rightarrow \text{free} \\ 0, & \text{if ch}(i) \rightarrow \text{busy} \end{cases} \quad (5)$$

T_{off} is sum of all available channels from given number of channels N for considered time slots (T) from history. if number of primary users arrival is zero, then $T_{\text{off}} = N$, at the same time primary user occupied channel for all time slots (T) then $T_{\text{off}} = 0$. So T_{off} values are ranges from 0 to N , if values are nearer to N then channel has more availability.

$$\text{CAV} = \left(\frac{C}{L}\right) * \text{BOF} * \frac{1}{N} \left(\bigcup_{\text{currentch}}^N \left(\frac{S_{lp}}{S_{hp}} \right) \right) * T_{\text{off}} \quad (6)$$

where,

- C – Participated no. of CR
- L – Participated no. of PR
- BOF – Best order factor
- N – No. of channels
- S_{lp} – slots occupied by low priority user
- S_{hp} – slots occupied by high priority user
- T_{off} – number of unused time slots
- A . Best order factor (BOF)

This is a unique sub-parameter used in this article. It has been defined as availability of channels calculated based on channel usage pattern. Calculation of maximum number of slots available continuously is one of the key objectives of BOF function. One of the main parameter for Eq. (6) for defining overall availability of

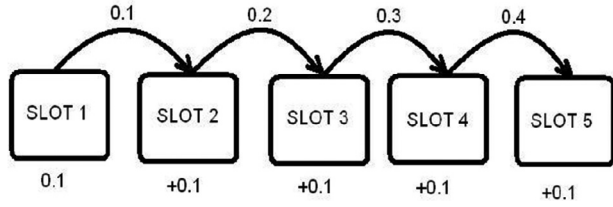


Fig. 6. Session value calculation (for single session).

Table 2
Rating values considered for AHP.

Comparison	Equal	Moderate	Strong	Very Strong	Extreme
Rating	1	3	5	7	9

channels. BOF will increase the opportunity to get optimal channel for long time transmission. Some of the channels may have empty slots, though it has the frequent usage of primary or cognitive. If this in case, after some short time interval, newly occupied SU has shifted again to others. Channel shifting or channel handoff will take a considerable amount of time and energy from each transmitting CR node.

$$BOF = \frac{1}{n} \sum_{slot.no(n)=0}^N \left(\frac{session_value_n}{count_n} \right) \quad (7)$$

where,

n – No. of sessions occurred in channel;
 N – Total No. of time slots in given history

In some cases, channels may have equal number of T_{off} values, which means equal number of free slots for two or more participating channels. Existing channel decision techniques handles this scenario by weighting those channels with equal importance. But the actual pattern of usage not considered by them, which will create a considerable impact in the channel decision model. In this research, it has been addressed and BOF parameter used along with the usual availability calculation. In BOF calculations, sessions are one of the simple and optimal ways of counting free slots continuity based on that total session value. Session value has to be calculated for considered N channels with T time slot details. The number of the session can range from 1 to $N/2$, if there is no PU activity then number of session is 1. Alternatively, each time slots PU activity found then number of sessions will be maximum $N/2$.

B. Session value calculation

In Fig. 5, there are three sessions has occurred with free slots in the fourth channel. Initially, every individual time slot having default continuity vector (0.1). Our aim is to detect long utilization space channels, so going to give higher values for continues free time slots, means long sessions. Increasing range from 0.1 to ascending order-up-to maximum value, then we have to find average. Like this every session value for each channel has to be calculated and updated to channel table for finding CDF value of that particular channel. Fig. 6 shows the sequence of counting continuity vector. The session value calculation showed as follows,

Here, we are going to do BOF calculation for channel 4 and the above diagram for session_2 calculation. It has five number of time slots continuously from slot1 to slot5. At SLOT5 total session value defined like this $(0.1 + 0.2 + 0.3 + 0.4 + 0.5) = 1.5$. Next step is to find average for that session value. Total session value divided by count of free time slots i.e., $(1.5/5) = 0.3$. Now apply this in (7),

Table 3
Priority order of performance parameters.

Text	PL > SNR > P > CAV > B > R
Voice	SNR > P > B > PL > CAV > R
Video	R > B > SNR > CAV > PL > P

Table 4
Comparative rating Matrix “A”.

	CAV	B	PL	S	SNR	(R)
CAV	1	4/3	4/9	4/6	4/7	4/2
B	3/4	1	3/9	3/6	3/7	3/2
PL	9/4	9/3	1	9/6	9/7	9/2
S	6/4	6/3	6/9	1	6/7	6/2
SNR	7/4	7/3	7/9	7/6	1	7/2
R	2/4	2/3	2/9	2/6	2/7	1

Table 5
EFAHP algorithm.

n : total number of channels
 m : number of perf_factors
 $m!$: possible number of priority order
 k : recessive priority order
 q : number of training spell required
 Where, q – defined factors based on hardware limitation || Problem_nature,
 $1 < q < (m! - k)$

I. Weighting parameters

If: known service (text||audio||video)

Traffic: poiss_rand()

{Min: 0.05, max: 0.7}

Routine EFAHP: (t = 1, t < q, t++) {

Define {

Rating_levels (1–9);

Perf_factors(m);}

Generate: Comparison_matrix $m \times m$ as ‘A’;

Assign Rating_perf_factors (1 to m); {

Compare (Fx, Fy), $x, y \leq m$;

$A[i][j] = (Fx/Fy)$;

Calc_row_sum: $rSum(i) = \sum_{j=1}^m A^2[i][j]$;

Until(smaller_value(Norm(rSum(i)))) {

Normalized row_sum: $rSum(i) / \sum_{i=1}^m rSum(i)$;

End

Consistency checking:

priority_order(perf_factors):high → low(Norm(rSum(i)));

Consistency Ratio: $CR = CI/RI$; //validating decision

If (CR less than 0.10): Decision → Consistent

Else: Decision → In-Consistent

where, $CI = \frac{\gamma_{max} - m}{m - 1}$, RI = defined($m(2:10)$ → 0 to 1.51)

$\gamma_{max} = \sum_{j=1}^m \text{Norm.rSum}[j] \cdot \text{Norm.cSum}[j]^T$

End

II. Finding Priority order of parameters:

ch_wgt → Normalized row_sum;

Channel Decision Model (ch_wgt);

Note: TOPSIS-I has been found most fit model for CCRN

Initialization of best outcomes:

Let, BCT(n) → Best Computation time with n channels

WFR(n) → Worst fault ratio with n channels

BRSD(n) → Best relative standard deviation among n samples

BAT(n) → Best Achieved throughput with n channels

If (BCT(n) > comp_time(t))

BCT(n) = comp_time(t);

If (WFR(n) > fault_ratio(t))

WFR(n) = fault_ratio(t);

If (BRSD(n) > relative_standard_deviation(t))

BRSD(n) = relative_standard_deviation(t);

If (BAT(n) > achieved_throughput(t))

BAT(n) = achieved_throughput(t);

Else: Routine EFAHP: (t = 1, t < m!, t++)

Best_priority_order(q) = ch_wgt(t){BCT(n), WFR(n), BRSD(n), BAT(n)}

End

Table 6
TOPSIS algorithm.

TOPSIS Algorithm:

C – Criteria considered for evaluation, $C=\{c_1, c_2, \dots, c_m\}$

m – Number of criteria's involved

n – Number of alternatives

 V_{ij} – quantitative value of A_i with respect to C_j W_m – derived weight's of 'm' criteria's, $0 \leq W_m \leq 1$ D_{mat} – quantitative decision matrix Ψ – Normalized decision matrix D_{mat} , $0 \leq V_{ij} \leq 1$ Ω – Weighted decision matrix D_{mat} , $0 \leq V_{ij} \leq 1$ $\omega(i)$ – Weight of alternatives by TOPSIS**Stage – 1:**

i. Normalization of quantitative data matrix

Let, $D_{mat} = [V_{ij}]_{m \times n}$ $\Psi = [\text{Normalized}(D_{mat})_{ij}]_{m \times n}$ **if(C's objective is benefit)**

$$[\text{Normalized}(D_{mat})_{ij}]_{m \times n} = \frac{V_{ij}}{\sqrt{\sum_{i=1}^n V_{ij}^2}}, i = 1, 2, 3 \dots m, j = 1, 2, 3 \dots n$$

Else(C's objective is cost)

$$[\text{Normalized}(D_{mat})_{ij}]_{m \times n} = \frac{\frac{1}{V_{ij}}}{\sqrt{\sum_{i=1}^n \left(\frac{1}{V_{ij}}\right)^2}}, i = 1, 2, 3 \dots m, j = 1, 2, 3 \dots n$$

Endii. Calculation of weighted $[\text{Normalized}(D_{mat})_{ij}]_{m \times n}$

$$\text{Let, } \Omega = V_{ij} \times W_m = \begin{bmatrix} V_{11} * W_1 & \dots & V_{m1} * W_m \\ \vdots & \ddots & \vdots \\ V_{n1} * W_1 & \dots & V_{mn} * W_m \end{bmatrix}$$

Stage -2:

iii. Positive Ideal Solutions

 $PIS(j^*) = \text{Max}(\Omega_{ij^*})$, where $i=1,2,3,\dots,n, j=1,2,3,\dots,m$,

Negative Ideal Solutions

iv. Positive Ideal matrix

$$\tau(i,j)_{m \times n} = (\Omega_{ij} - PIS(j))^2, \text{ where } i = 1, 2, 3, \dots, n, j = 1, 2, 3, \dots, m,$$

Negative Ideal matrix

$$\tau(i,j)_{m \times n} = (\Omega_{ij} - NIS(j))^2, \text{ where } i = 1, 2, 3, \dots, n, j = 1, 2, 3, \dots, m,$$

 $S^+(i) = \text{rowSum}(\tau(i,j)_{m \times n})$, where $i = 1, 2, 3, \dots, n$ $S^-(i) = \text{rowSum}(\tau(i,j)_{m \times n})$, where $i = 1, 2, 3, \dots, n$

v. Calculating weight of alternatives

$$\omega(i) = \frac{S^-}{S^+ + S^-}, i = 1, 2, 3 \dots, n, 0 \leq \omega(i) \leq 1$$

Example 1: BOF for Channel 4

$$= \left(\frac{\frac{0.1+0.2}{2} + \frac{0.1+0.2+0.3+0.4+0.5}{5} + \frac{0.1+0.2+0.3}{3} \right) \\ = \left(\frac{0.15+0.3+0.2}{3} \right) \\ = 0.21$$

3.1.2. Bandwidth (MHz)

Channel Bandwidth may range from few KHz to more than 40 MHz, within India 7 MHz channels mostly used for CRN. Two or more continuous channels can combine and form one wide range of channel also possible. This should be maximum for multimedia type of data (Wei and Lin, 2008).

3.1.3. Probability of packet loss (PL) (%)

When we try to send data over wireless channels, packet loss at receiver side is obvious thing in experiments (Eze et al., 2018). We have recorded channels past PL and used for finding better channel using MCDM.

3.1.4. Signal power (dBm)

Available channel Frequency should be less than –100dBm at least. If signal power of channel more than –100dBm then that can be considered as occupied channel (Chen et al., 2008).

3.1.5. SNR(dB)

It should be greater value, more than 10 dB, 20 dB, etc. threshold SNR we can set based on our need (Smitha and Vinod, 2012).

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (8)$$

3.1.6. Maximum rate of transfer (bps)

With the available bandwidth, the maximum rate of transmission will be recorded as a parameter (Govindan and Jepsen, 2016; Kim and Shin, 2008). For text services it can be slower but in case of multimedia transmission, it should be one of the dominant factor.

3.2. Channel ranking using EFAHP-TOPSIS**3.2.1. Efficient Feedback analytic Hierarchy process**

Usually, AHP technique will be used to define exact order of weight for factors involving in decision-making problems. But, giving order of priority is one big challenge in a dynamic way. In existing approaches group preferences or individual will define the priority order before proceeding AHP. But in CRN every time if we define priority manually, it will not be support for dynamic environment like text, audio and video service over same network. If, 'm' number of factors then m! Possible priority orders are possible to fix, we can't try all priority order in training time due to hardware limitation (Kumar et al., 2017). So for making our proposed framework to support any type of service, we derived Efficient Feedback Analytic Hierarchy Process (EFAHP) in Table 6, through that, if user know what type of service then he can fix $q = (m! - k)$ number of trial or training for finding better priority order. For that historical data about channel will be recorded in each CR's cache, from that we can shortlist better priority decisions of better performance outcomes. Overall we planned to fix limited training instance for finalizing order of priority by EFAHP using limited number of trial and error method.

3.2.1.1. Implementation of EFAHP for different data services. Calculate the weight of some important quality parameters of each channel, quality parameters such as Channel Deciding Factor (CDF), Bandwidth, SNR and Interference Ratio. For strengthening the decision we need give more preference to channel which has a high important quality parameter. So, In AHP process initially, each channels rating value of Optimal Channel Selection (our problem) value will be assumed as 1. Then based on some post process (like getting some comparative scores from expert about each parameter) we can calculate the weight of parameters as follows,

A. Series of steps needed for applying AHP (Qualitative Analysis)

- Creating Comparative rating Matrix "A"
- Calculate Square Matrix i.e., A^2
- Find Normalized Row Sums

As per Table 5 algorithm for EFAHP, using priority order from Table 3 and saaty weights from Table 2, we have constructed comparative rating matrix A and illustrated in Table 4. Eq. (9) has been applied to a comparative matrix and calculated normalized matrix as in Table 7.

$$\text{Normalize}(A[i][j]) = \frac{A^{[i][j]}}{\sum_{i=1}^m A^{[i][j]}} \quad (9)$$

B. Calculation of Normalized rowSum

$$rSum(i) = \sum_{j=1}^m A^2[i][j] \quad (10)$$

where $i \rightarrow 1$ to m : row numbers

Table 7

Normalized weighted comparative matrix.

5.99	7.99	2.65	3.99	3.44	11.98
4.50	5.99	1.99	3.00	2.58	8.99
13.51	18.01	5.98	9.00	7.75	27.02
9.01	12.02	3.99	6.01	5.17	18.03
10.51	14.01	4.65	7.00	6.03	21.02
3.00	4.00	1.33	2.00	1.72	6.00

Table 8

Row sum of normalized weighted comparative matrix.

36.04
27.04
81.27
54.22
63.22
18.05

Table 9

Calculated weights for performance factors.

CAV (%)	0.1288
Bandwidth (MHz)	0.0966
Prob of pkt loss PL (%)	0.2904
sig_pow (dBm)	0.1938
SNR (dB)	0.2259
max_rate (bps)	0.0645

Eq. (10) has been applied to Table 7 and rSum calculated as in Table 8.

Our aim in this part is Normalizing the rSum(i) for finding the priority order. It can be calculated by Eq. (11).

$$\text{Normalized_rSum}(i) = \text{rSum}(i) / \text{total}(\text{rSum}(i)), \quad i = 1, 2, \dots, m \quad (11)$$

where

$$\text{Total}(\text{rSum}(i)) = \sum_{i=1}^m \text{rSum}(i) \quad (12)$$

At last, for verifying calculated weight of performance factors, consistency check should have to done. Consistency ratio (Crat) already been defined by Saaty, which is the ratio of consistency index and random index as in Eq. (13), random indices are all pre-defined by Saaty (1994) (See Table 9).

$$\text{Crat} = \frac{CI}{RI} \quad (13)$$

where,

CI – Consistency index in Eq. (14)

RI – Random index is displayed in Table 10.

$$CI = \frac{\lambda_{\max} - n}{RI} \quad (14)$$

where,

n – Number of factors involved in EFAHP

After applying Eqs. (13) and (14), performance factors of text, audio and video data service, the following Table 11 has been calculated and each Crat values are found that less than 0.1. It clearly

shows that computed results are all consistent and ready to use for TOPSIS.

Eq. (11), should be continued and calculated until getting smaller enough value for easy computation. Thus we have computed priority order of performance parameters of text data service. Usually this Normalized rSum(i) value between 0 → 1. Also it should satisfy the following Eq. (15).

$$\sum_{i=1}^m \text{Normalized_rSum}(i) = 1 \quad (15)$$

Order of priority based on the requirement has been given as input for EFAHP, now result values of Normalized rSum(i) shows exact importance or weight for further channel ranking process. So far EFAHP used qualitative analytic method to fix the weight of performance parameters, through that we can able to do a quantitative comparison between CUP_factors (Jayakumar and Janakiraman, 2016). Table 12 has results of Normalized rSum(i) calculated for text, audio and video data by Eq. (11). Though the same number of performance factors has been considered for EFAHP, different results came for three different types of data services. So the order of priority has to be fixed carefully in channel selection methods.

We have done some of the interpretation in Table 12, in text transmission service we know clearly packet loss ratio (PL) is most important factor and it should be very less. But, maximum rate of transmission (R) should be a dominant factor for video data transmission and noise should play a vital role in audio transmission. In audio service, SNR is 3 times weight than CAV. In video transmission R is 2 times weightier than CAV. It shows, though CAV should be important for all three services with equal importance, other factors decide the channel quality much more showed in Fig. 7.

3.2.2. Ranking channels using TOPSIS

To rank a set of alternatives, the TOPSIS method as outranking relation theory was used to analyze the data of a decision matrix. The Elimination and Choice Translating Reality (TOPSIS) method was first introduced in (Jayakumar et al., 2016). It is one of the most extensively used outranking methods reflecting the decision maker's preferences in many fields. The TOPSIS approach was then developed by a number of variants. The transport sustainability was firstly evaluated by TOPSIS method in (Jayakumar et al., 2016); then the modification of TOPSIS was used to reduce the subjectivity of decision makers. TOPSIS for best choice problem, it is suitable for channel selection, since some of the CR going to rise channel request, best choices enough to compute. Also we validated TOPSIS with other methods, it outperforms well.

3.2.2.1. Implementation of TOPSIS for different data services with EFAHP weights. Input for TOPSIS ranking method here is, n-number of Channels with their quality parameter weights (that we derived

Table 10

Random index by Saaty.

n	1	2	3	4	5	6	7	8
R	0.0	0.0	0.5	0.9	1.1	1.2	1.3	1.4

Table 11

Crat values calculated for sample criteria.

Service	Text	Audio	Video
CI	0.0000013	–1.159346	0.000000
Crat	0.0000011	–0.966121	0.000000
	Consistent	Consistent	Consistent

Table 12
Priority order derived by MAHP.

	Text_data	Audio_data	Video_data
CAV	0.1288	0.099	0.129
B	0.0966	0.1933	0.2258
PL	0.2904	0.1284	0.0968
P	0.1938	0.2253	0.0645
SNR	0.2259	0.2896	0.1935
R	0.0645	0.0643	0.2903

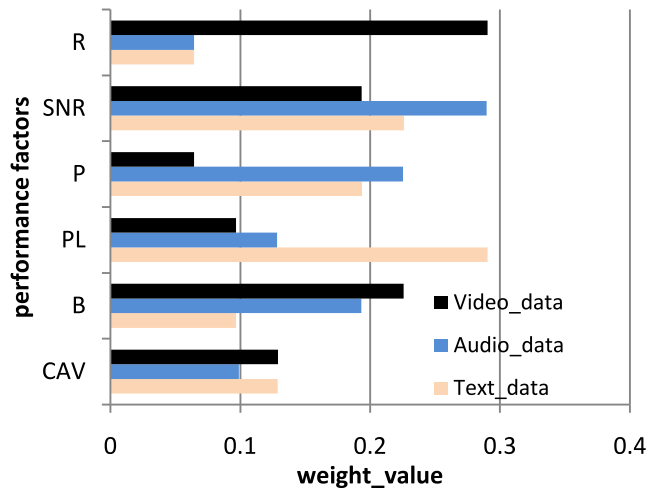


Fig. 7. performance factor weight comparison with multiple scenarios.

previously by EFAHP). Each alternative (i.e., channels) compared with each other and after n -number of comparison, channels will be getting some single weight value. Based on that rank table, the rank will be prepared by the CH.

Series of steps needed for applying TOPSIS (Quantitative Analysis)

- Collecting Each Channel Properties
- Normalize the Properties and make it as matrix “B”
- Ranking the Channels

Implementing the proposed model, we have considered 20, 100 and 200 channels in simulation. In Table 13, list of 20 channels and its past historical performance values on different parameters has been displayed. Since, values are in different ranges that should get normalized and weighted as in Tables 14, 17, and 20 for further process in TOPSIS as follows,

$$X_{ij} = A_{ij} / \sqrt{\sum_{i=1}^M A_{ij}^2} \quad (16)$$

Then, Positive Ideal Solution (PIS) and negative Ideal Solution (NIS) has been calculated in Table 14, and Positive Ideal Matrix (PIM) and Negative Ideal Matrix (NIM) as in Tables 15 and 16 has to be calculated for text data transmission as in stage II steps in Table 6.

In the same way, Positive Ideal Solution (PIS) and negative Ideal Solution (NIS) has been calculated in Table 17, and Positive Ideal Matrix (PIM) and Negative Ideal Matrix (NIM) as in Tables 18 and 19 has to be calculated for text data transmission as in stage II steps in Table 6.

In the same way, Positive Ideal Solution (PIS) and negative Ideal Solution (NIS) has been calculated in Table 20, and Positive Ideal Matrix (PIM) and Negative Ideal Matrix (NIM) as in Tables 21 and 22 has to be calculated for text data transmission as in stage II steps in Table 6.

Table 13
Considered channel properties for simulation.

Ch	CAV	B	PL	S	SNR	(R)
1	0.75	20	10	−106	26	172
2	0.41	20	20	−114	35	95
3	0.91	10	10	−129	45	130
4	0.45	40	26	−128	22	66
5	0.81	20	5	−92	28	179
6	0.51	20	16	−128	12	114
7	0.53	40	32	−88	32	45
8	0.4	40	17	−125	11	163
9	0.67	20	18	−80	38	155
10	0.76	20	21	−97	13	123
11	0.7	20	15	−120	12	155
12	0.78	20	12	−107	21	133
13	0.7	20	22	−107	22	59
14	0.3	20	13	−129	14	118
15	0.5	20	23	−122	12	107
16	0.5	20	17	−107	25	85
17	0.8	10	10	−105	26	193
18	0.45	40	28	−128	41	156
19	0.4	20	6	−125	31	101
20	0.5	20	5	−112	18	148

Table 14
Text: normalized weighted matrix.

	0.129 CAV	0.097 B	0.290 PL	0.194 P	0.226 SNR	0.064 R
c1	0.035	0.017	0.068	−0.045	0.050	0.019
c2	0.019	0.017	0.034	−0.042	0.067	0.010
c3	0.043	0.009	0.068	−0.037	0.087	0.014
c4	0.021	0.035	0.026	−0.037	0.042	0.007
c5	0.038	0.017	0.136	−0.051	0.054	0.020
c6	0.024	0.017	0.043	−0.037	0.023	0.013
c7	0.025	0.035	0.021	−0.054	0.062	0.005
c8	0.019	0.035	0.040	−0.038	0.021	0.018
c9	0.031	0.017	0.038	−0.059	0.073	0.017
c10	0.036	0.017	0.032	−0.049	0.025	0.014
c11	0.033	0.017	0.045	−0.039	0.023	0.017
c12	0.036	0.017	0.057	−0.044	0.040	0.015
c13	0.033	0.017	0.031	−0.044	0.042	0.006
c14	0.014	0.017	0.052	−0.037	0.027	0.013
c15	0.023	0.017	0.030	−0.039	0.023	0.012
c16	0.023	0.017	0.040	−0.044	0.048	0.009
c17	0.037	0.009	0.068	−0.045	0.050	0.021
c18	0.021	0.035	0.024	−0.037	0.079	0.017
c19	0.019	0.017	0.114	−0.038	0.060	0.011
c20	0.023	0.017	0.136	−0.042	0.035	0.016
PIS	0.043	0.035	0.136	−0.037	0.087	0.021
NIS	0.014	0.009	0.021	−0.059	0.021	0.005

In Table 23, we have illustrated rank of 20 channels calculated from EFAHP-TOPSIS for three different types of data transmission such as text, audio and video and Fig. 8 shows the same. Since parameter importance will change for each scenario, the rank of channels will be different for scenarios. Eg. Channel 5 will be best for text transmission but for audio 4th importance and 2nd importance for video transmission. Channel 20 is 6th best for video transmission but not fit for text transmission. So need-based channel selection can be possible when we set the required order of parameter before channel selection. This can be tuned dynamically by using our proposed EFAHP. Feedback will be collected periodical time interval, and weights of parameters will be adjusted and re-rank will be calculated.

3.3. Validating EFAHP-TOPSIS ranking results

Validation of weights calculated from the proposed model has been compared with weights calculated from other two parallel MCDM techniques such as SAW and MOORA. Anyhow parameter

Table 15

Text: positive ideal matrix.

CAV	B	PL	P	SNR	R	S* (rSUM)
0.0001	0.0003	0.0046	0.0001	0.0013	0.0000	0.080
0.0005	0.0003	0.0105	0.0000	0.0004	0.0001	0.109
0.0000	0.0007	0.0046	0.0000	0.0000	0.0000	0.073
0.0005	0.0000	0.0121	0.0000	0.0020	0.0002	0.121
0.0000	0.0003	0.0000	0.0002	0.0011	0.0000	0.040
0.0004	0.0003	0.0088	0.0000	0.0040	0.0001	0.116
0.0003	0.0000	0.0132	0.0003	0.0006	0.0003	0.121
0.0006	0.0000	0.0093	0.0000	0.0043	0.0000	0.119
0.0001	0.0003	0.0097	0.0005	0.0002	0.0000	0.104
0.0000	0.0003	0.0108	0.0001	0.0038	0.0001	0.123
0.0001	0.0003	0.0083	0.0000	0.0040	0.0000	0.113
0.0000	0.0003	0.0063	0.0001	0.0021	0.0000	0.094
0.0001	0.0003	0.0111	0.0001	0.0020	0.0002	0.117
0.0008	0.0003	0.0070	0.0000	0.0036	0.0001	0.109
0.0004	0.0003	0.0114	0.0000	0.0040	0.0001	0.127
0.0004	0.0003	0.0093	0.0001	0.0015	0.0001	0.108
0.0000	0.0007	0.0046	0.0001	0.0013	0.0000	0.082
0.0005	0.0000	0.0125	0.0000	0.0001	0.0000	0.114
0.0006	0.0003	0.0005	0.0000	0.0007	0.0001	0.047
0.0004	0.0003	0.0000	0.0000	0.0027	0.0000	0.059

Table 16

Text: negative ideal matrix.

CAV	B	PL	P	SNR	R	S-(rSUM)
0.0004	0.0001	0.0022	0.0002	0.0008	0.0002	0.063
0.0000	0.0001	0.0002	0.0003	0.0021	0.0000	0.052
0.0008	0.0000	0.0022	0.0005	0.0043	0.0001	0.089
0.0000	0.0007	0.0000	0.0005	0.0004	0.0000	0.041
0.0006	0.0001	0.0132	0.0001	0.0011	0.0002	0.123
0.0001	0.0001	0.0005	0.0005	0.0000	0.0001	0.034
0.0001	0.0007	0.0000	0.0000	0.0016	0.0000	0.050
0.0000	0.0007	0.0004	0.0005	0.0000	0.0002	0.041
0.0003	0.0001	0.0003	0.0000	0.0027	0.0001	0.059
0.0005	0.0001	0.0001	0.0001	0.0000	0.0001	0.029
0.0004	0.0001	0.0006	0.0004	0.0000	0.0001	0.039
0.0005	0.0001	0.0013	0.0002	0.0004	0.0001	0.050
0.0004	0.0001	0.0001	0.0002	0.0004	0.0000	0.035
0.0000	0.0001	0.0010	0.0005	0.0000	0.0001	0.041
0.0001	0.0001	0.0001	0.0004	0.0000	0.0000	0.026
0.0001	0.0001	0.0004	0.0002	0.0007	0.0000	0.039
0.0005	0.0000	0.0022	0.0002	0.0008	0.0003	0.064
0.0000	0.0007	0.0000	0.0005	0.0033	0.0001	0.069
0.0000	0.0001	0.0085	0.0005	0.0015	0.0000	0.103
0.0001	0.0001	0.0132	0.0003	0.0002	0.0001	0.118

Table 17

Audio: normalized weighted matrix.

	0.099 CAV	0.193 B	0.128 PL	0.225 P	0.290 SNR	0.064 R
c1	0.027	0.035	0.030	-0.052	0.064	0.019
c2	0.015	0.035	0.015	-0.048	0.086	0.010
c3	0.033	0.017	0.030	-0.043	0.111	0.014
c4	0.016	0.070	0.012	-0.043	0.054	0.007
c5	0.029	0.035	0.060	-0.060	0.069	0.020
c6	0.018	0.035	0.019	-0.043	0.030	0.013
c7	0.019	0.070	0.009	-0.063	0.079	0.005
c8	0.014	0.070	0.018	-0.044	0.027	0.018
c9	0.024	0.035	0.017	-0.069	0.094	0.017
c10	0.027	0.035	0.014	-0.057	0.032	0.013
c11	0.025	0.035	0.020	-0.046	0.030	0.017
c12	0.028	0.035	0.025	-0.051	0.052	0.015
c13	0.025	0.035	0.014	-0.051	0.054	0.006
c14	0.011	0.035	0.023	-0.043	0.035	0.013
c15	0.018	0.035	0.013	-0.045	0.030	0.012
c16	0.018	0.035	0.018	-0.051	0.062	0.009
c17	0.029	0.017	0.030	-0.052	0.064	0.021
c18	0.016	0.070	0.011	-0.043	0.101	0.017
c19	0.014	0.035	0.050	-0.044	0.077	0.011
c20	0.018	0.035	0.060	-0.049	0.044	0.016
PIS	0.033	0.070	0.060	-0.043	0.111	0.021
NIS	0.011	0.017	0.009	-0.069	0.027	0.005

Table 18

Audio: positive ideal matrix.

CAV	B	PL	P	SNR	R	S*(rSUM)
0.0000	0.0012	0.0009	0.0001	0.0022	0.0000	0.067
0.0003	0.0012	0.0020	0.0000	0.0006	0.0001	0.066
0.0000	0.0028	0.0009	0.0000	0.0000	0.0000	0.061
0.0003	0.0000	0.0024	0.0000	0.0032	0.0002	0.078
0.0000	0.0012	0.0000	0.0003	0.0018	0.0000	0.057
0.0002	0.0012	0.0017	0.0000	0.0066	0.0001	0.099
0.0002	0.0000	0.0026	0.0004	0.0010	0.0003	0.067
0.0003	0.0000	0.0018	0.0000	0.0070	0.0000	0.096
0.0001	0.0012	0.0019	0.0007	0.0003	0.0000	0.065
0.0000	0.0012	0.0021	0.0002	0.0062	0.0001	0.099
0.0001	0.0012	0.0016	0.0000	0.0066	0.0000	0.098
0.0000	0.0012	0.0012	0.0001	0.0035	0.0000	0.078
0.0001	0.0012	0.0022	0.0001	0.0032	0.0002	0.083
0.0005	0.0012	0.0014	0.0000	0.0059	0.0001	0.095
0.0002	0.0012	0.0022	0.0000	0.0066	0.0001	0.102
0.0002	0.0012	0.0018	0.0001	0.0024	0.0001	0.077
0.0000	0.0028	0.0009	0.0001	0.0022	0.0000	0.077
0.0003	0.0000	0.0025	0.0000	0.0001	0.0000	0.053
0.0003	0.0012	0.0001	0.0000	0.0012	0.0001	0.054
0.0002	0.0012	0.0000	0.0000	0.0044	0.0000	0.077

Table 19

Audio: negative ideal matrix.

CAV	B	PL	P	SNR	R	S-(rSUM)
0.0003	0.0003	0.0004	0.0003	0.0014	0.0002	0.053
0.0000	0.0003	0.0000	0.0004	0.0035	0.0000	0.066
0.0005	0.0000	0.0004	0.0007	0.0070	0.0001	0.093
0.0000	0.0028	0.0000	0.0007	0.0007	0.0000	0.065
0.0003	0.0003	0.0026	0.0001	0.0018	0.0002	0.073
0.0001	0.0003	0.0001	0.0007	0.0000	0.0001	0.034
0.0001	0.0028	0.0000	0.0000	0.0027	0.0000	0.074
0.0000	0.0028	0.0001	0.0006	0.0000	0.0002	0.060
0.0002	0.0003	0.0001	0.0000	0.0044	0.0001	0.072
0.0003	0.0003	0.0000	0.0001	0.0000	0.0001	0.029
0.0002	0.0003	0.0001	0.0005	0.0000	0.0001	0.036
0.0003	0.0003	0.0002	0.0003	0.0006	0.0001	0.043
0.0002	0.0003	0.0000	0.0003	0.0007	0.0000	0.040
0.0000	0.0003	0.0002	0.0007	0.0001	0.0001	0.036
0.0001	0.0003	0.0000	0.0006	0.0000	0.0000	0.031
0.0001	0.0003	0.0001	0.0003	0.0012	0.0000	0.044
0.0003	0.0000	0.0004	0.0003	0.0014	0.0003	0.052
0.0000	0.0028	0.0000	0.0007	0.0055	0.0001	0.095
0.0000	0.0003	0.0017	0.0006	0.0024	0.0000	0.071
0.0001	0.0003	0.0026	0.0004	0.0003	0.0001	0.061

Table 20

Video: normalized weighted matrix.

	0.129 CAV	0.226 B	0.097 PL	0.065 P	0.194 SNR	0.290 R
c1	0.035	0.041	0.023	-0.015	0.043	0.085
c2	0.019	0.041	0.011	-0.014	0.058	0.047
c3	0.043	0.020	0.023	-0.012	0.074	0.064
c4	0.021	0.082	0.009	-0.012	0.036	0.033
c5	0.038	0.041	0.045	-0.017	0.046	0.089
c6	0.024	0.041	0.014	-0.012	0.020	0.056
c7	0.025	0.082	0.007	-0.018	0.053	0.022
c8	0.019	0.082	0.013	-0.013	0.018	0.081
c9	0.031	0.041	0.013	-0.020	0.063	0.077
c10	0.036	0.041	0.011	-0.016	0.021	0.061
c11	0.033	0.041	0.015	-0.013	0.020	0.077
c12	0.037	0.041	0.019	-0.015	0.035	0.066
c13	0.033	0.041	0.010	-0.015	0.036	0.029
c14	0.014	0.041	0.017	-0.012	0.023	0.058
c15	0.023	0.041	0.010	-0.013	0.020	0.053
c16	0.023	0.041	0.013	-0.015	0.041	0.042
c17	0.038	0.020	0.023	-0.015	0.043	0.096
c18	0.021	0.082	0.008	-0.012	0.068	0.077
c19	0.019	0.041	0.038	-0.013	0.051	0.050
c20	0.023	0.041	0.045	-0.014	0.030	0.073
PIS	0.043	0.082	0.045	-0.012	0.074	0.096
NIS	0.014	0.020	0.007	-0.020	0.018	0.022

Table 21

Video: positive ideal matrix.

CAV	B	PL	P	SNR	R	S* (rSUM)
0.0001	0.0017	0.0005	0.0000	0.0010	0.0001	0.003
0.0005	0.0017	0.0012	0.0000	0.0003	0.0024	0.006
0.0000	0.0038	0.0005	0.0000	0.0000	0.0010	0.005
0.0005	0.0000	0.0013	0.0000	0.0014	0.0040	0.007
0.0000	0.0017	0.0000	0.0000	0.0008	0.0000	0.003
0.0004	0.0017	0.0010	0.0000	0.0030	0.0015	0.007
0.0003	0.0000	0.0015	0.0000	0.0005	0.0054	0.008
0.0006	0.0000	0.0010	0.0000	0.0031	0.0002	0.005
0.0001	0.0017	0.0011	0.0001	0.0001	0.0004	0.003
0.0000	0.0017	0.0012	0.0000	0.0028	0.0012	0.007
0.0001	0.0017	0.0009	0.0000	0.0030	0.0004	0.006
0.0000	0.0017	0.0007	0.0000	0.0016	0.0009	0.005
0.0001	0.0017	0.0012	0.0000	0.0014	0.0044	0.009
0.0008	0.0017	0.0008	0.0000	0.0026	0.0014	0.007
0.0004	0.0017	0.0013	0.0000	0.0030	0.0018	0.008
0.0004	0.0017	0.0010	0.0000	0.0011	0.0029	0.007
0.0000	0.0038	0.0005	0.0000	0.0010	0.0000	0.005
0.0005	0.0000	0.0014	0.0000	0.0000	0.0003	0.002
0.0006	0.0017	0.0001	0.0000	0.0005	0.0021	0.005
0.0004	0.0017	0.0000	0.0000	0.0020	0.0005	0.005

Table 22

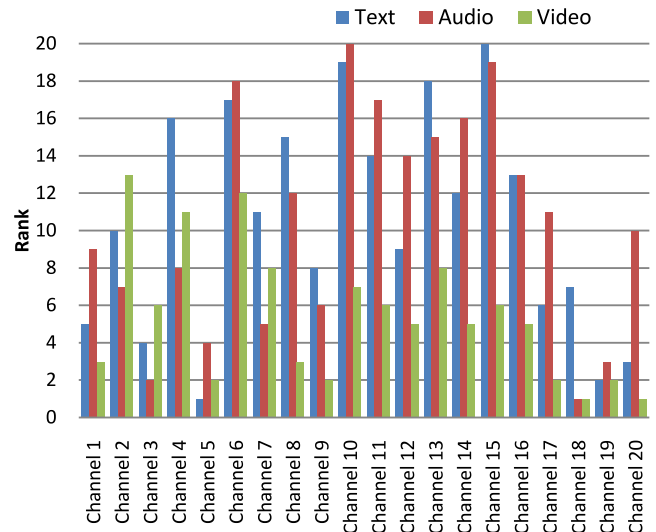
Video: negative Ideal matrix.

CAV	B	PL	P	SNR	R	S-(rSUM)
0.0004	0.0004	0.0002	0.0000	0.0006	0.0040	0.006
0.0000	0.0004	0.0000	0.0000	0.0016	0.0006	0.003
0.0008	0.0000	0.0002	0.0001	0.0031	0.0018	0.006
0.0000	0.0038	0.0000	0.0001	0.0003	0.0001	0.004
0.0006	0.0004	0.0015	0.0000	0.0008	0.0044	0.008
0.0001	0.0004	0.0001	0.0001	0.0000	0.0012	0.002
0.0001	0.0038	0.0000	0.0000	0.0012	0.0000	0.005
0.0000	0.0038	0.0000	0.0001	0.0000	0.0034	0.007
0.0003	0.0004	0.0000	0.0000	0.0020	0.0030	0.006
0.0005	0.0004	0.0000	0.0000	0.0000	0.0015	0.002
0.0004	0.0004	0.0001	0.0000	0.0000	0.0030	0.004
0.0005	0.0004	0.0001	0.0000	0.0003	0.0019	0.003
0.0004	0.0004	0.0000	0.0000	0.0003	0.0000	0.001
0.0000	0.0004	0.0001	0.0001	0.0000	0.0013	0.002
0.0001	0.0004	0.0000	0.0000	0.0000	0.0009	0.002
0.0001	0.0004	0.0000	0.0000	0.0005	0.0004	0.001
0.0005	0.0000	0.0002	0.0000	0.0006	0.0054	0.007
0.0000	0.0038	0.0000	0.0001	0.0024	0.0030	0.009
0.0000	0.0004	0.0009	0.0001	0.0011	0.0008	0.003
0.0001	0.0004	0.0015	0.0000	0.0001	0.0026	0.005

Table 23

Channel ranking for text, audio and video by using EFAHP-TOPSIS (20 Channels).

Channels	$S-/(S^* + S-)$ wgt	Rank	$S-/(S^* + S-)$ wgt	Rank	$S-/(S^* + S-)$ wgt	Rank
Channel 1	0.440	5	0.444	9	0.630	3
Channel 2	0.325	10	0.499	7	0.308	13
Channel 3	0.548	4	0.605	2	0.535	6
Channel 4	0.254	16	0.454	8	0.374	11
Channel 5	0.754	1	0.559	4	0.750	2
Channel 6	0.228	17	0.257	18	0.193	12
Channel 7	0.290	11	0.527	5	0.399	8
Channel 8	0.257	15	0.385	12	0.595	3
Channel 9	0.362	8	0.525	6	0.625	2
Channel 10	0.192	19	0.227	20	0.258	7
Channel 11	0.259	14	0.270	17	0.390	6
Channel 12	0.348	9	0.355	14	0.401	5
Channel 13	0.228	18	0.322	15	0.118	8
Channel 14	0.272	12	0.275	16	0.208	5
Channel 15	0.172	20	0.235	19	0.157	6
Channel 16	0.263	13	0.364	13	0.176	5
Channel 17	0.436	6	0.400	11	0.562	2
Channel 18	0.375	7	0.641	1	0.807	1
Channel 19	0.686	2	0.567	3	0.402	2
Channel 20	0.669	3	0.443	10	0.512	1

**Fig. 8.** Channel Ranking for Text, Audio and Video by using EFAHP-TOPSIS with 20 channels.**Table 24**

Text: validating EFAHP-TOPSIS ranking results from EFAHP-TOPSIS with EFAHP-SAW and EFAHP-MOORA by using Relative Standard Deviation (RSD).

Channels	SAW	MOORA	TOPSIS
Channel 1	0.145077354	0.165295011	0.439838032
Channel 2	0.107073921	0.146011656	0.325081017
Channel 3	0.183732178	0.13301259	0.547568476
Channel 4	0.094905322	0.125812234	0.253928777
Channel 5	0.213884699	0.121329055	0.754110321
Channel 6	0.082629468	0.116490923	0.22786091
Channel 7	0.093866178	0.10643621	0.290404492
Channel 8	0.095050139	0.104369652	0.256758722
Channel 9	0.117754158	0.099557342	0.36235503
Channel 10	0.075279403	0.085575722	0.192440932
Channel 11	0.09640289	0.081881189	0.258626399
Channel 12	0.121622854	0.078959629	0.347647546
Channel 13	0.085844764	0.077635662	0.227701229
Channel 14	0.087213066	0.073635309	0.272177375
Channel 15	0.066605445	0.067915889	0.171996601
Channel 16	0.094234859	0.060335785	0.263427838
Channel 17	0.14055106	0.060237779	0.435832919
Channel 18	0.139502772	0.052560024	0.375338755
Channel 19	0.182756416	0.043988866	0.685892302
Channel 20	0.185908026	0.03901476	0.668832002
AVG	0.120495	0.092003	0.367891
RSD	35.315396	37.907311	46.541987

weight calculation scheme fixed as EFAHP for all three. Most common statistical comparative technique Relative Standard Deviation (RSD) has used for validation. It shows how far results are varied from mean value of weights. If RSD value is high, then we can state calculated weights are consistent and concrete decision can be possible. One more statistical model ANOVA (single factor) has been used to find significant difference in the calculated results. From this we can observe differences in the techniques and check whether these differences will effect significantly in the channel decision or not. Tables 24, 26, and 28 clearly shows that EFAHP-TOPSIS gives better RSD than other competitive techniques. As far ANOVA test Tables 25, 27, and 29 proves significant difference between channel selection techniques. i.e., null hypothesis been rejected for all three scenarios. We have applied this ranking model for 100 and 200 numbers of channels also with varying PU traffic. Results of all 20, 100 and 200 channels are applied under RSD and

Table 25

ANOVA: text.

SUMMARY						
Groups	Count	Sum	Average	Variance		
SAW	20	2.409895	0.120495	0.001811		
MOORA	20	1.840055	0.092003	0.001216		
TOPSIS	20	7.35782	0.367891	0.029318		
ANOVA						
Source of variation	SS	df	MS	F	P-value	F crit
Between groups	0.920873	2	0.460437	42.70595	4.64E–12	3.158843
Within groups	0.614549	57	0.010782			
Total	1.535422	59				

Table 26

Audio: validating EFAHP-TOPSIS ranking results from EFAHP-TOPSIS with EFAHP-SAW and EFAHP-MOORA by using Relative Standard Deviation (RSD).

Channels	SAW	MOORA	TOPSIS
Channel 1	0.12323922	0.176073924	0.444217478
Channel 2	0.113344829	0.165136839	0.498949907
Channel 3	0.163043539	0.216841005	0.605197147
Channel 4	0.116312137	0.162905533	0.454149043
Channel 5	0.153354032	0.185730727	0.558884715
Channel 6	0.071308005	0.126683594	0.257097998
Channel 7	0.119850231	0.160769896	0.527173833
Channel 8	0.10310928	0.157689283	0.385369896
Channel 9	0.117839484	0.176572286	0.524974662
Channel 10	0.065527996	0.117341912	0.226681358
Channel 11	0.081028748	0.13606271	0.269768746
Channel 12	0.103160599	0.157773854	0.355169645
Channel 13	0.0832143	0.13321869	0.322034391
Channel 14	0.073811213	0.129757972	0.275147178
Channel 15	0.062330418	0.111668818	0.235314973
Channel 16	0.090306761	0.144295251	0.364377408
Channel 17	0.109349556	0.162234395	0.400014761
Channel 18	0.172247681	0.216463346	0.641283472
Channel 19	0.143197808	0.182893113	0.566976116
Channel 20	0.124804216	0.155385446	0.442878803
AVG	0.110	0.159	0.418
RSD	29.151	18.218	30.826

Table 27

ANOVA: audio.

SUMMARY						
Groups	Count	Sum	Average	Variance		
SAW	20	2.19038	0.109519	0.001019		
MOORA	20	3.175499	0.158775	0.000837		
TOPSIS	20	8.355662	0.417783	0.016586		
ANOVA						
Source of variation	SS	df	MS	F	P-value	F crit
Between groups	1.096921	2	0.54846	89.21946	2.78E–18	3.158843
Within groups	0.350397	57	0.006147			
Total	1.447318	59				

ANOVA test. Table 30, shows clearly better RSD values for all three scenarios even increase in the PU traffic for all three different types of data services. Table 31, shows clearly rejection of null hypothesis for most of the scenarios even increase in the PU traffic for all three different types of data services.

4. Experimental analysis

In addition to statistical validation method for proposed model, we have decided to perform simulated experiments also. Actually, we have tried real-time experiments with USRP devices before simulation as in Fig. 9. Though USRP CR is capable to sense

600 MHz to 6 GHz, in our geo-location (16.2334°N, 80.5509°E) most of the ISM bands were free. With more number of free bands (PU_traffic = 0) and are very less (3 No's) number of USRP device, our proposed model not necessary to implement. When at-least PU_traffic > 0.2 and 20 numbers of channels, proposed model can be implemented for better management of spectrum. Finally, we have decided to simulate by using MATLAB, setup for simulating with various scenarios has been designed using GUI.

As simulation inputs, number of channels (20, 100, 200), number of CR nodes (10, 50, 100), type of data service (Text, Audio, Video), simulation time (600, 1200, 2400 s), channel selection technique (EFAHP-TOPSIS, EFAHP-SAW, EFAHP-MOORA), size of

Table 28

Video: validating EFAHP-TOPSIS ranking results from EFAHP-TOPSIS with EFAHP-SAW and EFAHP-MOORA by using Relative Standard Deviation (RSD).

Channels	SAW	MOORA	TOPSIS
Channel 1	0.21190279	0.202642386	0.63015367
Channel 2	0.162383751	0.174499377	0.308127306
Channel 3	0.212160905	0.197323599	0.534659967
Channel 4	0.168244035	0.186934634	0.374161802
Channel 5	0.241934559	0.207960279	0.749852562
Channel 6	0.142893418	0.144044939	0.192770511
Channel 7	0.1708598	0.209120745	0.399266143
Channel 8	0.200093874	0.203965584	0.594622712
Channel 9	0.204601613	0.223240208	0.62495639
Channel 10	0.153401509	0.171851378	0.258247068
Channel 11	0.172217361	0.173053083	0.390360832
Channel 12	0.182114276	0.178789692	0.401003282
Channel 13	0.134785067	0.152143865	0.117872307
Channel 14	0.141698902	0.135727686	0.20832884
Channel 15	0.134033403	0.14932976	0.156911249
Channel 16	0.146261513	0.154544129	0.175533096
Channel 17	0.20405455	0.195062962	0.561985902
Channel 18	0.243497002	0.265227317	0.80665118
Channel 19	0.186008561	0.152094296	0.401565659
Channel 20	0.198625912	0.159048322	0.511629318
AVG	0.181	0.182	0.420
RSD	18.553	17.546	48.130

Table 29

ANOVA: video.

SUMMARY						
Groups	Count	Sum	Average	Variance		
SAW	20	3.611773	0.180589	0.001123		
MOORA	20	3.636604	0.18183	0.001018		
TOPSIS	20	8.39866	0.419933	0.04085		
ANOVA						
Source of variation	SS	df	MS	F	P-value	F crit
Between groups	0.759868	2	0.379934	26.51269	7.24E-09	3.158843
Within groups	0.816825	57	0.01433			
Total	1.576693	59				

Table 30

Overall Relative Standard Deviation (RSD) comparison between EFAHP-TOPSIS, EFAHP-SAW and EFAHP-MOORA.

PU traffic	Service type	Relative Standard Deviation (RSD)								
		EFAHP-TOPSIS			EFAHP-SAW			EFAHP-MOORA		
		No. of channels 20	No. of channels 100	No. of channels 200	20	100	200	20	100	200
0.2	Text	46.529	67.32	45.02	35.30	58.3	3.372	37.12	34.02	7.466
	Audio	30.74	68.06	29.26	29.26	7.766	26.74	18.67	3.518	5.541
	Video	48.06	64.53	52.67	18.74	54.07	46.02	17.06	21.04	6.704
0.4	Text	66.53	78.54	51.65	20.98	75.95	54.72	33.09	7.263	16.71
	Audio	86.02	70.39	55.48	80.22	63.25	43.6	5.005	23.81	55.93
	Video	63.6	70.42	69.41	73.99	1.537	72.18	16.5	65.74	62.99
0.6	Text	48.19	74.475	79.34	7.647	68.15	64.74	31.02	17.59	49.24
	Audio	35.27	79.58	63.98	7.948	4.051	28.4	15.35	67.35	58.37
	Video	84.44	72.79	42.32	26.31	73.33	20.05	68.92	46.37	33.33

data (10, 100, 200 MB), time interval between sensing (5, 10, 20 secs), Initial energy of CR 500 J and traffic_type = poisson_random. Basic experimental block design is illustrated in Fig. 10.

4.1. Performance Analysis

4.1.1. Computation time

The computation time based analyses on different MCDM based channel management techniques show that the channel manage-

ment techniques with EFAHP-TOPSIS at channel selection level outperform other channel management technique EFAHP-MOORA, but compared to EFAHP-SAW and Random selection techniques taken less time for channel selection. Since it is independent of node size and sensing techniques, any type of channel sensing model with EFAHP-TOPSIS will give same order of results as in Table 32. Even though Random and EFAHP-SAW methods have given less average computation time than other methods, selected channel quality factor will be unpredictable. EFAHP-TOPSIS

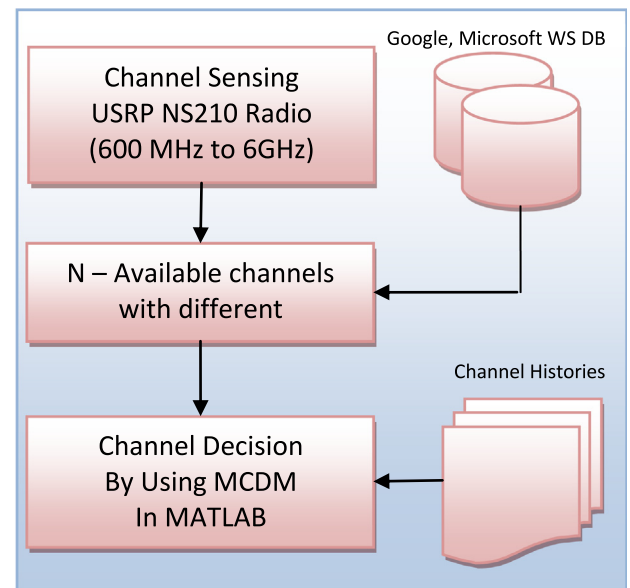
Table 31

Overall ANOVA Test results comparison between EFAHP-TOPSIS, EFAHP-SAW and EFAHP-MOORA.

PU traffic	Service type	Anova (Ho rejected) status		
		No. of channels = 20 Sig. Diff	No. of channels = 100 Sig. Diff	No. of channels = 200 Sig. Diff
0.2	Text	✓	✓	X
	Audio	✓	✓	✓
	Video	✓	✓	✓
0.4	Text	✓	✓	X
	Audio	✓	✓	✓
	Video	✓	X	✓
0.6	Text	✓	✓	✓
	Audio	✓	✓	✓
	Video	X	✓	✓

**Fig. 9.** USRP N210 SDR used for lab experimental setup.

effectively considers historical values about each channel based on that decision had been taken, so it may take additional time than others but quality of channels can be predicted and matched with CR's request effectively. Table 32 shows Best, Worst and Average computation, i.e. same number of channels applied under selection technique in multiple instances and calculated best, worst and average as illustrated in Figs. 11–13.

**Fig. 10.** Experimental functional design.**Table 32**

Analyses of Computation time (msecs).

No. CH	Random			EFAHP-SAW			EFAHP – TOPSIS			EFAHP – MOORA		
	Best	Worst	Avg	Best	Worst	Avg	Best	Worst	Avg	Best	Worst	Avg
5	1.4	2.9	2.1	9.3	10.8	10.2	34.2	35.2	35.0	37.6	38.7	38.0
10	3.4	4.8	3.8	12.3	13.4	12.8	43.0	44.7	43.8	49.2	49.8	49.8
20	5.3	5.6	5.4	15.7	16.3	15.9	54.0	54.9	54.7	65.0	65.8	65.2
30	8.2	8.7	8.4	19.6	20.2	19.9	67.0	68.0	67.3	85.1	85.8	85.4
40	9.4	10.9	10.1	24.8	25.2	24.9	82.5	83.6	82.7	106.3	107.7	106.8
50	14.2	15.3	15.0	30.4	32.0	31.1	105.4	106.4	105.9	132.8	134.1	133.5
60	26.2	27.8	27.0	38.7	39.4	38.9	135.2	136.1	135.6	166.3	167.2	166.9
70	31.4	32.9	32.0	48.2	49.0	48.6	173.2	173.5	173.5	207.9	209.2	208.6
80	44.0	45.1	45.0	60.7	61.4	60.8	221.9	222.3	222.1	266.1	267.5	267.0
90	51.5	55.5	53.5	74.0	77.0	76.0	283.3	285.3	284.3	340.7	342.7	341.7
100	58.0	69.0	63.0	89.0	98.0	95.0	358.9	369.9	363.9	433.4	438.4	437.4
110	68.5	76.5	72.5	115.7	121.7	118.7	463.8	467.8	465.8	553.9	564.9	559.9
120	81.0	86.0	82.0	145.4	153.4	148.4	575.2	581.2	576.2	635.6	642.6	641.6
130	86.5	96.5	91.5	179.5	189.5	185.5	628.8	637.8	633.8	715.0	722.0	718.0
140	100.0	103.0	101.0	230.9	234.9	231.9	694.2	701.2	697.2	837.1	843.1	840.1
150	94.5	111.5	110.5	279.9	304.9	289.9	763.9	775.9	766.9	972.9	988.9	982.9
160	112.0	127.0	120.0	353.4	367.4	362.4	829.6	853.6	843.6	1144.0	1159.0	1150.0
170	116.5	131.5	129.5	447.0	454.0	453.0	918.9	930.9	927.9	1343.5	1353.5	1345.5
180	134.0	145.0	139.0	550.2	579.2	566.2	1056.1	1082.1	1067.1	1402.8	1418.8	1412.8
190	136.5	156.5	148.5	699.8	722.8	707.8	1219.2	1242.2	1227.2	1476.4	1492.4	1483.4
200	145.0	169.0	158.0	874.7	893.7	884.7	1408.3	1414.3	1411.3	1549.6	1573.6	1557.6

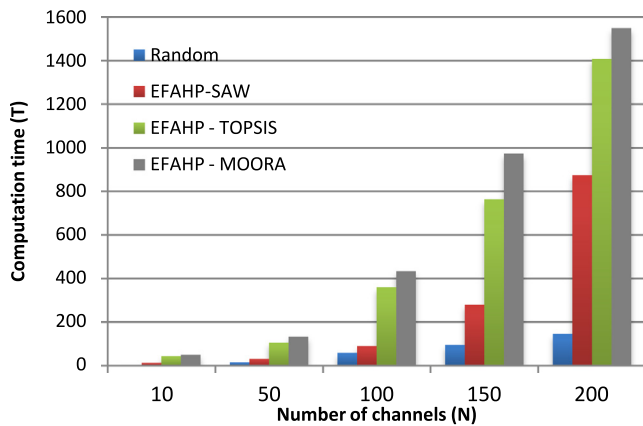


Fig. 11. Computation time (Best case).

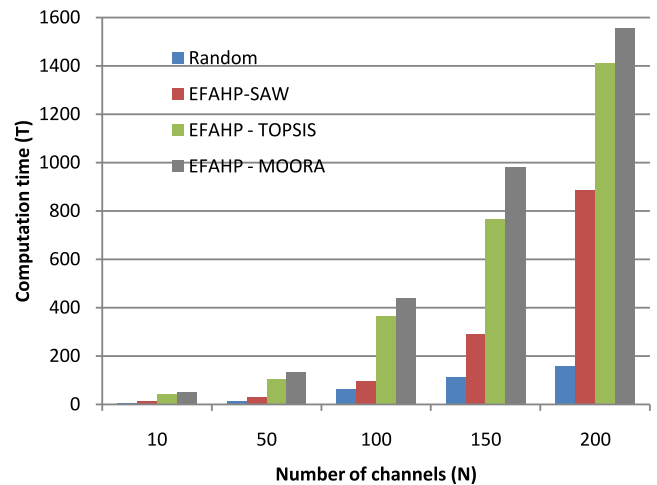


Fig. 13. Computation time (Avg case).

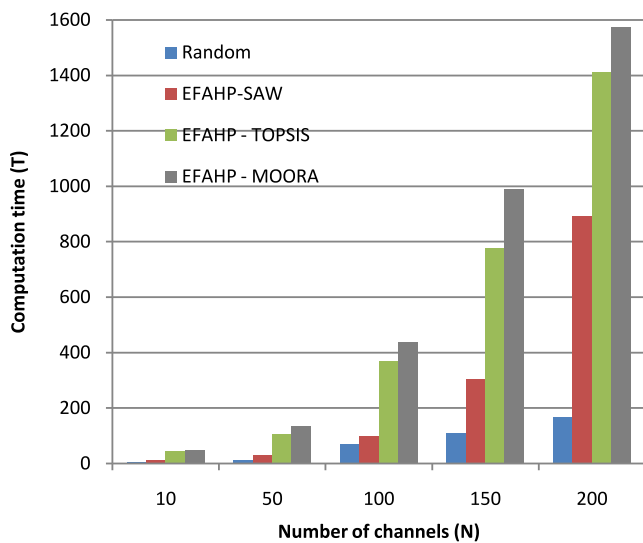


Fig. 12. Computation time (Worst case).

4.1.2. Number of handoffs

The handoff from the current operating band to a new free band is costlier operation in CRN. Number of handoff's influence much in overall CRN performances and it is directly proportional to end-to-end delay, energy consumption, packet loss, and throughput. In this simulation, we have assumed fixed data size 100 MB and the calculated number of handoffs for various channel selection techniques. As we expected EFAHP-TOPSIS outperforms well compared to other techniques. Though "random" (Poisson random) selection process will take very less computation time, a number of handoffs from "random" method is high compared to other techniques. Since channels activity or availability not been considered in the

selection, randomly it decides to switch over on-going PU transmission channel, which will create unnecessary interference and leads channel handoff. But other channel selection techniques such as EFAHP-SAW, EFAHP-MOORA and EFAHP-TOPSIS has considered historical performance details about CRN, so that selection can find right free channel to continue new/interrupted data transmission.

Particularly, EFAHP-TOPSIS shows less number of handoffs. Though the computation time is high for proposed model than EFAHP-SAW and EFAHP-MOORA, one of most important performance evaluation number of handoffs gave better result for cooperative CRN as in Table 33. Since number of handoffs direct proportional to delay and energy consumption, results under same scenario also similar to this. So, we have illustrated only handoffs results under different circumstances in Figs. 14 and 15. If PU arrival rate or PU traffic increases, obviously probability of interferences also will be high then handoff is the only solution to continue transmission. We have considered low traffic as PU_traffic = 0.2 (20% of the PU users as active in CRN) and high traffic as PU_traffic = 0.6 (60% of the PU users as active in CRN).

5. Conclusion

An integrated channel selection scheme has been designed and adopted from operation research area. Two famous and classical models such as AHP and TOPSIS has been integrated to detect free-channels. Instead of considering AHP in its original form, feedback supported EFAHP has been integrated with TOPSIS for better channel detection. Instead of considering only band_size of a channel, multiple factors are considered to evaluate the channel capability. Based on the requirement or need of CRN system EFAHP can tune the parameter weight through that better ranking of

Table 33
Number of handoffs for varying traffic, channel size and data service.

PU traffic	Service type	Number of spectrum handoff's											
		EFAHP-TOPSIS No. of channels			EFAHP-SAW No. of channels			EFAHP – MOORA No. of channels			Random No. of channels		
		20	100	200	20	100	200	20	100	200	20	100	200
Low traffic (0.2)	Text	246	213	198	313	272	252	290	252	234	447	388	360
	Audio	283	245	228	360	312	290	334	290	269	514	446	414
	Video	372	327	304	419	367	340	390	341	315	593	516	480
High traffic (0.6)	Text	493	412	389	627	524	495	582	487	460	896	749	707
	Audio	557	466	439	709	592	559	658	550	519	1012	846	799
	Video	830	697	657	934	785	741	870	730	689	1330	1114	1053

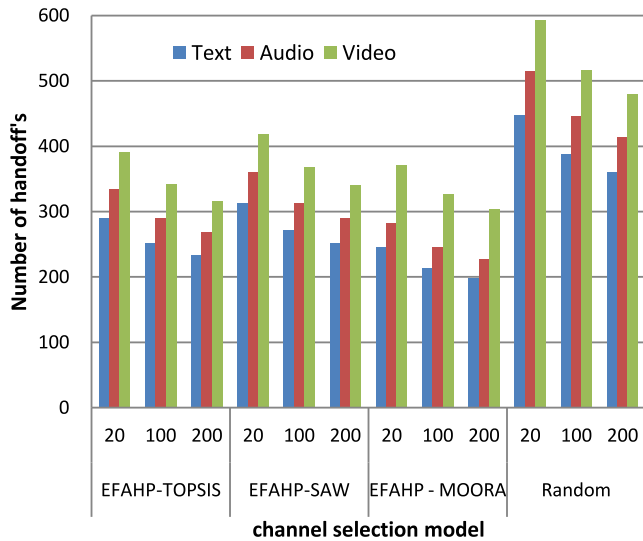


Fig. 14. Number of handoffs vs Number of channels with various channel selection schemes under PU Traffic = 0.2 (low).

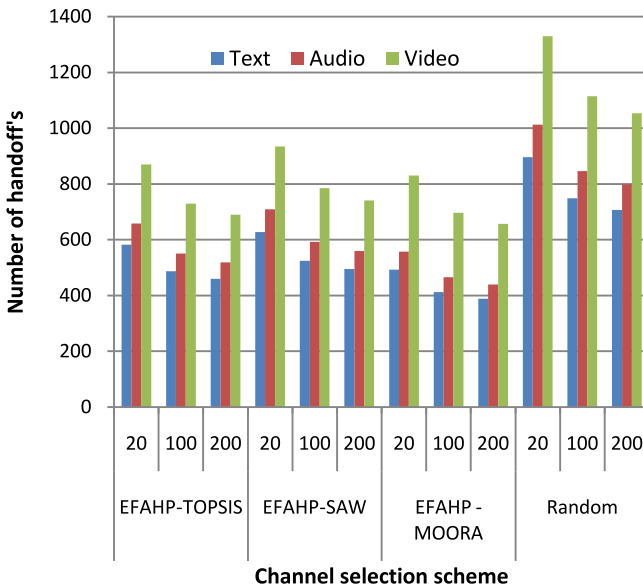


Fig. 15. Number of handoffs vs Number of channels with various channel selection schemes under PU Traffic = 0.6 (high).

channels is possible. We defined algorithm designed and simulated though standard tool. Initially, we have validated the results from EFAHP-TOPSIS with other parallel possible channel selection schemes such as EFAHP-SAW and EFAHP-MOORA. Finally, we have done performance analyses in computation time and number of handoffs with varying size of channel, different data service and PU traffic rate and proved our proposed model outperforms well than others. In future, we can attach some additional factors of channels like, best time interval between each successive sense, Auction based on trustworthiness of participating nodes in cooperative CRN to improve QoS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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